Jacob Chalk

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Research Focus

My research focus is on multi-modal video understanding, particularly audio-visual. Here, my aim is to leverage both audio and video modalities to their maximum potential for solving challenging problems such as action recognition and detection, particularly in egocentric videos. More recently, I have also begun exploring video understanding in 3D. Specifically, I am investigating how eye-gaze primes object interaction and how we can leverage this to solve challenging 2D video/3D scene problems.

Education

PhD in Computer Vision *September 2021 – Current*
University of Bristol *Bristol, UK*

- *Current Thesis Title: Audio-Visual Video Understanding*
- *Anticipated Graduation Date: September 2025 (09/25)*

MEng in Computer Science *September 2017 – September 2021*
University of Bristol *Bristol, UK*

- *Degree Classification: First Class Honours*
- *Dissertation Title: Video GANS for Human-Object Interactions (83%)*

Teaching

Teaching Assistant *September 2019 – September 2024*
University of Bristol *Bristol, UK*
Software Product Engineering (2nd Year Module), Games Project (3rd Year Module), Computer Graphics (3rd Year Module) Image Processing and Computer Vision (3rd Year Module), Applied Deep Learning (4th Year Module)

Publications

- *Perrett, T.*, Darkhalil, A.*, Sinha, S.*, Emara, O.*, Pollard, S.*, Parida, K.*, Liu, K.*, Gatti, P.*, Bansal, S.*, Flanagan, K.*, Chalk, J.*, Zhu, Z.*, Guerrier, R.*, Abdelazim, F.*, Zhu, B., Moltisanti, D., Wray, M., Doughty, H., and Damen, D. 2025. HD-EPIC: A Highly-Detailed Egocentric Video Dataset.* arXiv preprint, arXiv:2502.04144. [Project Webpage] | [Paper] | [Code]
- *Plizzari, C., Goel, S., Perrett, T., Chalk, J., Kanazawa, A. and Damen, D., 2025. Spatial Cognition from Egocentric Video: Out of Sight, Not Out of Mind. International Conference on 3D Vision (3DV)* [Project Webpage] | [Paper] | [Code]
- *Chalk, J.*, Huh, J.*, Kazakos, E., Zisserman, A. and Damen, D., 2024. TIM: A Time Interval Machine for Audio-Visual Action Recognition. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (pp. 18153-18163)* [Project Webpage] | [Paper] | [Code]
- *Huh, J.*, Chalk, J.*, Kazakos, E., Damen, D. and Zisserman, A., 2023, June. Epic-sounds: A large-scale dataset of actions that sound. In ICASSP 2023-2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP) (pp. 1-5). IEEE.* [Project Webpage] | [Paper] | [Code]

*: Equal Contribution

Datasets

HD-EPIC *Released February 2025*
Highly Detailed Egocentric Video Dataset
HD-EPIC is an extensive dataset manually annotated with highly detailed and interconnected ground-truth labels covering: recipe steps, fine-grained actions, ingredients with nutritional values, moving objects, and audio annotations. All annotations are grounded in 3D through digital twinning of the scene, fixtures, object locations, and primed with gaze. Footage is collected from unscripted recordings in diverse home environments, making HD-EPIC the first dataset collected in-the-wild but with detailed annotations matching those in controlled lab environments. The dataset consists of 41 hours of video in 9 kitchens with digital twins of 413 kitchen fixtures,

capturing 69 recipes, 59K fine-grained actions, 51K audio events, 20K object movements and 37K object masks lifted to 3D. On average, there are 263 annotations per minute across the unscripted videos. HD-EPIC also presents a challenging VQA benchmark of 26K questions assessing the capability to recognise recipes, ingredients, nutrition, fine-grained actions, 3D perception, object motion, and gaze direction.

Roles:

- Gathered audio annotations matching a similar process to EPIC-Sounds (see below)
- Computed and visualised eye-gaze priming statistics for all 3D objects start and end locations. This involved projecting the eye-gaze direction ray into the 3D environment and measuring when the ray intersected with the objects 3D-bounding box before it was pick-up, or the end locations bounding box when placed down.
- Created eye-gaze related VQA questions, including *"What is the camera wearer currently looking at?"*, which involved finding the intersection between the projected gaze ray and 3D environment, as well as *"What will the person interact with next?"* which involved using objects whose start locations were primed with eye-gaze before being picked up to indicate it was about to be moved by the camera wearer.
- Ran LLM inference of Llama 3.2 (Vision-Instruct-90B) on the VQA questions

EPIC-Sounds *Released January 2023*
Audio-Centric Annotated Dataset *Paper Published in ICASSP 2023*
EPIC-Sounds is an audio-only dataset, gathered across 100 hours of untrimmed audios from the videos of EPIC-KITCHENS. This dataset was motivated by the observation that audio-visual networks tend to use both temporally and semantically identical annotations for both modalities, yet audio and video can significantly differ even when describing the same action.

Roles:

- Post-processed the labelled segments from annotators, this included:
  - Manually correcting typos in the free-form text descriptions.
  - Removing erroneous/redundant annotations.
  - Categorising the free-form descriptions into the 44 classes in the dataset.
  - Decomposing challenging segments containing multiple overlapping sounds into distinct annotations.
  - Training binary classifiers on the head classes of the dataset to clean the training set.
- Trained and ran quantitative analysis on the baseline models.
- Visualisation of the class distribution for the dataset.
- Ran analyses on the gathered data, such as the duration distribution of all classes and interplay between the visual and audio labels.
- Set and currently manage two challenges on the dataset: Audio-Based Interaction Recognition and Audio-Based Interaction Detection

Honours and Awards

EgoVis 2022/23 Distinguished Paper Awards *June 2024*
First Joint Egocentric Vision Workshop *IEEE/CVF Conference (CVPR)*
Awarded to our 2023 ICASSP paper: "EPIC-Sounds: A large-scale dataset of actions that sound."

EPIC-KITCHENS Challenges Winner *June 2024*
First Joint Egocentric Vision Workshop *IEEE/CVF Conference (CVPR)*
Audio-Based Interaction Recognition (2nd), Audio-Based Interaction Detection (2nd), Action Detection (3rd)

Top 5 Third Year MEng Computer Science/Computer Science with Maths Student *October 2020*
Netcraft *University of Bristol*

Research Activities

Presentations

- EPIC-KITCHENS Challenges. Presented at the First Joint Egocentric Vision Workshop (CVPR 2024).
- Oral Presentation for EPIC-Sounds. Presented at ICASSP 2023.

Reviewing

- 2025: CVPR, IJCV
- 2024: ECCV, TPAMI

Technical Knowledge and Programming

Programming Languages

- Python (Strongest)
- C++ (Intermediate)
- JavaScript (Intermediate)

Frameworks

- PyTorch (Strongest)
- TensorFlow (Intermediate)

Technical Fields

- Machine Learning, particularly Deep Learning
- Computer Vision
- Data Science